[Total No. of Questions :3] [2511]-1303101 Seat No.: [redacted]

Society for Computer Technology and Research's
PUNE INSTITUTE OF COMPUTER TECHNOLOGY
IN-SEM EXAMINATION OCT-2025
(S.Y.) (B.Tech.) (CE) (SEM-III)
(1303101) (Data Structures)
(2024 PATTERN)

[Time : 1 Hours] [Max Marks : 20]

Instructions to the candidates:

1. *Assume suitable data wherever required.*
2. *Neat diagrams must be drawn wherever necessary.*
3. *Use of non-programmable calculators is allowed.*
4. *[Marks], [Course Outcome] and [Bloom's Taxonomy Level] given at the end of each questions.*
5. ***Attempt Any Two Questions***

Q.1)A) Define Time and Space complexity of algorithm in terms of Big O, Omega and Theta Notation.

```
findFactorial(number):

// INPUT

 // number = A positive integer

// OUTPUT

 // Returns the factorial of the number

 if number = 1: return 1

else: return number * findFactorial(number - 1)

end findFactorial
```

Derive the time and space complexity of the above algorithm. [05] [CO1][L3]

B) Write the pseudocode of sentinel search. Analyze the time and space complexity using frequency count. Find the number of comparisons to find the element 50 in the list 20,30, 40, 15, 50, 60, 10, 20 using sentinel search. [05] [CO1][L4]

Q.2)A) Write the pseudocode for the sorting method of O(nlogn) time complexity for all cases. Give the output of each iterations for the given

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list to be sorted using the same algorithm. 10, 20, 15, 30, 5, 48, 9, 8. [06] [CO1][L3]

B) What is an associative array? How to find the element in an associative array? Give with an example. [04] [CO1][L2]

Q.3)A) Write the pseudocode for quick sort and apply and show the output of each iteration on elements 10, 60, 50, 30 20. [04] [CO1] [L3]

B) Compute the iterations of Binary search for the elements 20, 30, 40, 50, 70 and searching element 20. Derive its time complexity. [04] [CO1] [L3]

C) What is collision and overflow in Hashing. When do collision and overflow occur together? [02] [CO1] [L2]

····· End of question paper·····

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Total No. of Questions :3] [2512]-1303102 Seat No.:

Society for Computer Technology and Research's
PUNE INSTITUTE OF COMPUTER TECHNOLOGY
IN-SEM EXAMINATION OCT-2025
(SY) (B.Tech.) (CE) (SEM-III)
(1303102) (Computer Organisation and Architecture)
(2024 PATTERN)

Time : 1 Hours] [Max Marks : 20

Instructions to the candidates:

1. *Assume suitable data wherever required.*
2. *Neat diagrams must be drawn wherever necessary.*
3. *Use of non-programmable calculators is allowed.*
4. *[Marks], [Course Outcome] and [Bloom's Taxonomy Level] given at the end of each questions.*
5. ***Attempt Any Two Questions***

Q.1)A) Draw and explain instruction pipelining with an example. [4 Marks][CO1][L2]

B) Analyze the following assembly code segment and determine the final contents of the AX=BEh and CX= CBh registers, along with the status of the conditional flag bits (CF, ZF, SF) after execution. [6 Marks][CO2][L4]

1. ADDL $A4, %%Ax

2.ROL $3, %%Cx

3.NOT %%Ax

Q.2)A) Convert 0.265 decimal number into the IEEE 754 32-bit single-precision representation. Show all intermediate steps (binary conversion, normalization, exponent biasing, mantissa formation, and final representation in binary) [4 Marks][CO1][L3]

B) Analyze the effect of branch control instructions on instruction flow and stack. [6 Marks][CO6][L4]

Q.3)A) Explain characteristics of RISC architecture [4 Marks][CO1][L2]

B) Write a C program using inline assembly to count the number of occurrences of the character 'a' in a string terminated with '$' [6 Marks][CO2][L3]

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Total No. of Questions :3] [2511]-1303103 Seat No.: [redacted]

Society for Computer Technology and Research's
PUNE INSTITUTE OF COMPUTER TECHNOLOGY
IN-SEM EXAMINATION OCT-2025
(SE) (B.Tech.) (CE) (SEM-III)
(1303103) (Discrete Mathematics)
(2024 PATTERN)

Time : 1 Hours **Max Marks : 20**

Instructions to the candidates:

1. *Assume suitable data wherever required.*
2. *Neat diagrams must be drawn wherever necessary.*
3. *Use of non-programmable calculators is allowed.*
4. *[Marks], [Course Outcome] and [Bloom's Taxonomy Level] given at the end of each questions.*
5. ***Attempt Any Two Questions***

Q.1) A] In a survey of 60 people: 25 read newsweek magazine 26 read time, 26 read fortune, 9 read both newsweek and fortune, 11 read both newsweek and time, 8 read both time and fortune, 8 read no magazine at all. Fill in the correct number in all the regions of the Venn diagram.

i) Find the no. of people who read all three magazines.

ii) Find the no. of people who read exactly one magazine. [03][CO1][Apply]

B] A = {1,2,3} B={1,2,3,4,5} Find P(AUB) and P(A^B) [03][CO1][Apply]

C] Five pencils and 5 pens are to be arranged in a row. In how many ways they can be arranged if

i) All pencils must be arranged together

ii) No two pencils should be kept together and

iii) One pen and one pencil must be arranged together? [04][CO1][Analysis]

Q.2) A] A committee of 5 people is to be chosen from a group of 6 men and 4 women. How many committees are possible if there are no restrictions?

i)There are to be 3 men and 2 women?

ii)There are to be men only?

iii)There is to be a majority of women? [03][CO1][Analysis]

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B] A zip code contains 6 digits. How many different zipcodes can be made with the digits 0-9 if

i) No digit is used more than once

ii)First digit is not zero [03][CO1][Apply]

C] Use mathematical induction to show for natural no. n

$1^3+2^3+3^3+.....+n^3=(1+2+3+.....+n)^2$ [04][CO1][Apply]

Q.3) A] If 4 Maths books are selected from 6 different Maths books and 3 English books are chosen from 5 different English books, how many ways can the seven books be arranged on a shelf

i) If there are no restrictions?

ii) Maths book is at the beginning of the shelf?

iii) English is at the beginning and an Maths book is in the middle of the shelf. [04][CO1][Analysis]

B] Show that $\sim(p \vee (\sim p \wedge q))$ and $(\sim p \wedge \sim q)$ are logically equivalent using truth table. [03][CO1][Apply]

C] State the converse, contrapositive, and inverse of each of these conditional statements. [03][CO1][Apply]

i) I go to the beach whenever it is a sunny summer day.

ii) When I stay up late, it is necessary that I sleep until noon.

····· End of question paper·····

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